



RANDOM WALKS ON UNIPOTENT GROUPS

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OBJECTIVE

We consider the group G of unipotent matrices of the form

$$\begin{pmatrix} I_m & X & Z \\ 0 & I_n & Y \\ 0 & 0 & I_l \end{pmatrix}$$

where $X \in \mathcal{M}_{m \times n}(\mathbb{F}_q)$, $Y \in \mathcal{M}_{n \times l}(\mathbb{F}_q)$, and $Z \in \mathcal{M}_{m \times l}(\mathbb{F}_q)$ under matrix multiplication. We partition our group into equivalence classes, or superclasses, defined on the right.

Objective: Determine the probability of moving a matrix $M \in G$ from one superclass to another by multiplying on the right by a matrix $N \in G$ where $\text{rank}(N - I)$ is 0 or 1.

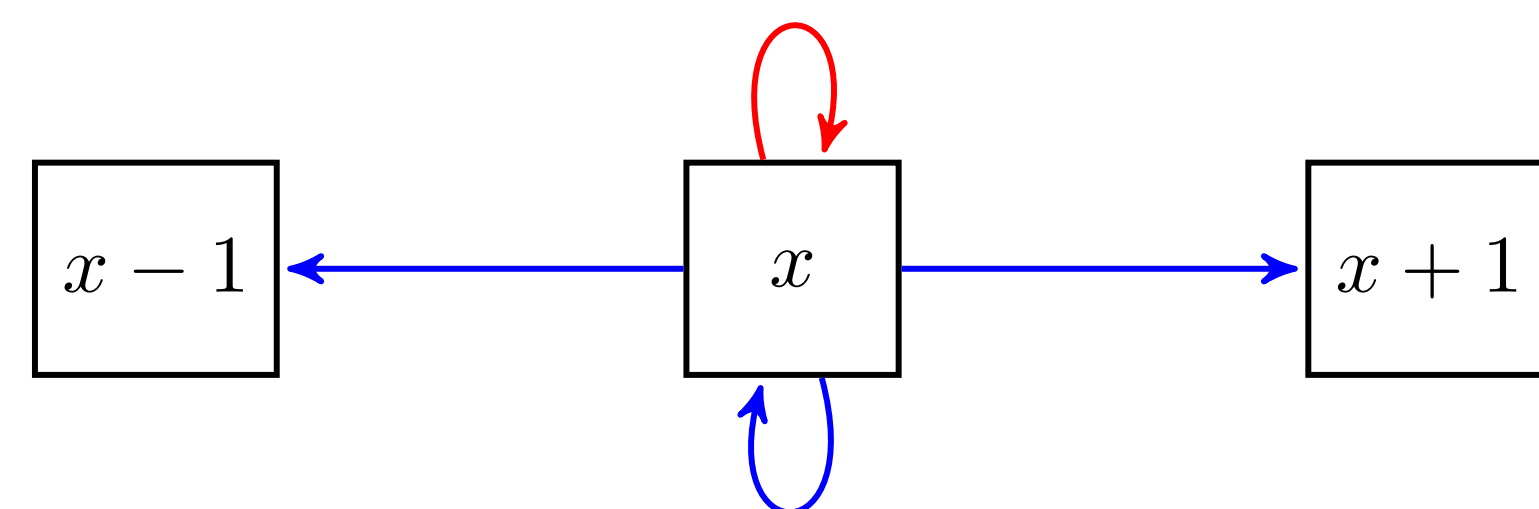
1-BLOCK CASE

- We start by considering the group of matrices of the form

$$\begin{pmatrix} I_m & X \\ 0 & I_n \end{pmatrix}$$

where $X \in \mathcal{M}_{m \times n}(\mathbb{F}_q)$.

- Partition this group by the following equivalence relation: two matrices M and N are equivalent if $\text{rank}(M - I) = \text{rank}(N - I)$.
- Let M be a matrix in our group and $x = \text{rank}(M - I)$. When M is multiplied on the right by a matrix N with $\text{rank}(N - I)$ is 0 or 1 then the rank of MN is in $\{x \pm \{0, 1\}\}$ depicted in the diagram below.



- The number of paths corresponding to these arrows are

$$x - 1 : q^{x-1}[x]$$

$$x \text{ (red)} : 1$$

$$x \text{ (blue)} : (q^m + q^n)[x] - q^{x-1}[x+1] - [2x-1]$$

$$x + 1 : q^{2x}(q-1)[m-x][n-x]$$

$$\text{where } [a] = \frac{q^a - 1}{q - 1}.$$

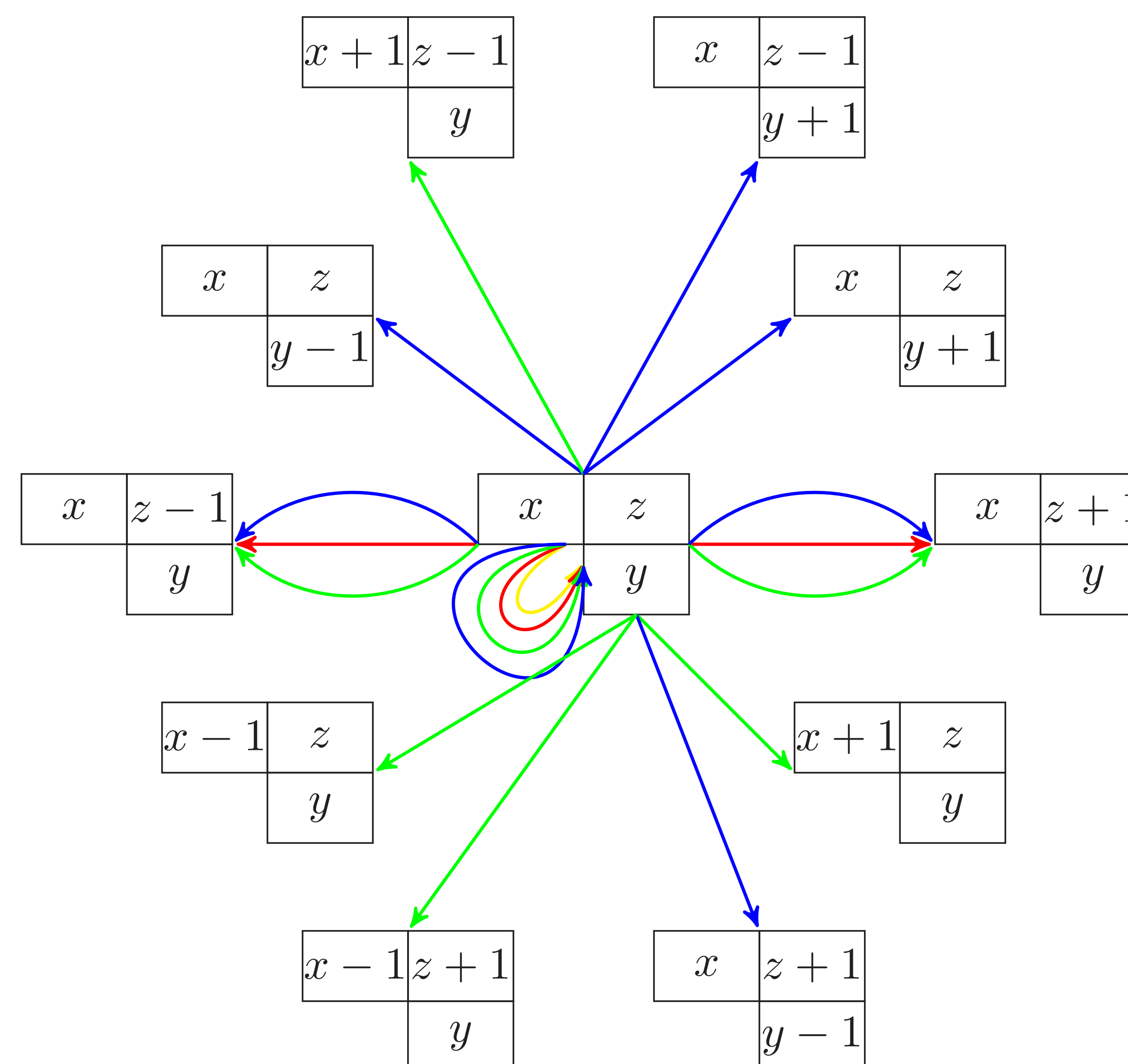
3-BLOCK CASE

- We partition our group G into superclasses given by the following equivalence relation.

$$M = \begin{pmatrix} I_m & X & Z \\ 0 & I_n & Y \\ 0 & 0 & I_l \end{pmatrix} \text{ and } N = \begin{pmatrix} I_m & A & C \\ 0 & I_n & B \\ 0 & 0 & I_l \end{pmatrix} \text{ are in the equivalence class } \begin{pmatrix} x & z \\ & y \end{pmatrix},$$

where $x = \text{rank}(X)$, $y = \text{rank}(Y)$, and $z = \text{rank}(M - I) - \text{rank}(X) - \text{rank}(Y)$, if

- $\text{rank}(X) = \text{rank}(A)$,
 - $\text{rank}(Y) = \text{rank}(B)$, and
 - $\text{rank}(M - I) - \text{rank}(X) - \text{rank}(Y) = \text{rank}(N - I) - \text{rank}(A) - \text{rank}(B)$.
- The diagram shows the superclasses that a matrix can move to when multiplied by a matrix N in G where $\text{rank}(N - I)$ equals 0 or 1.



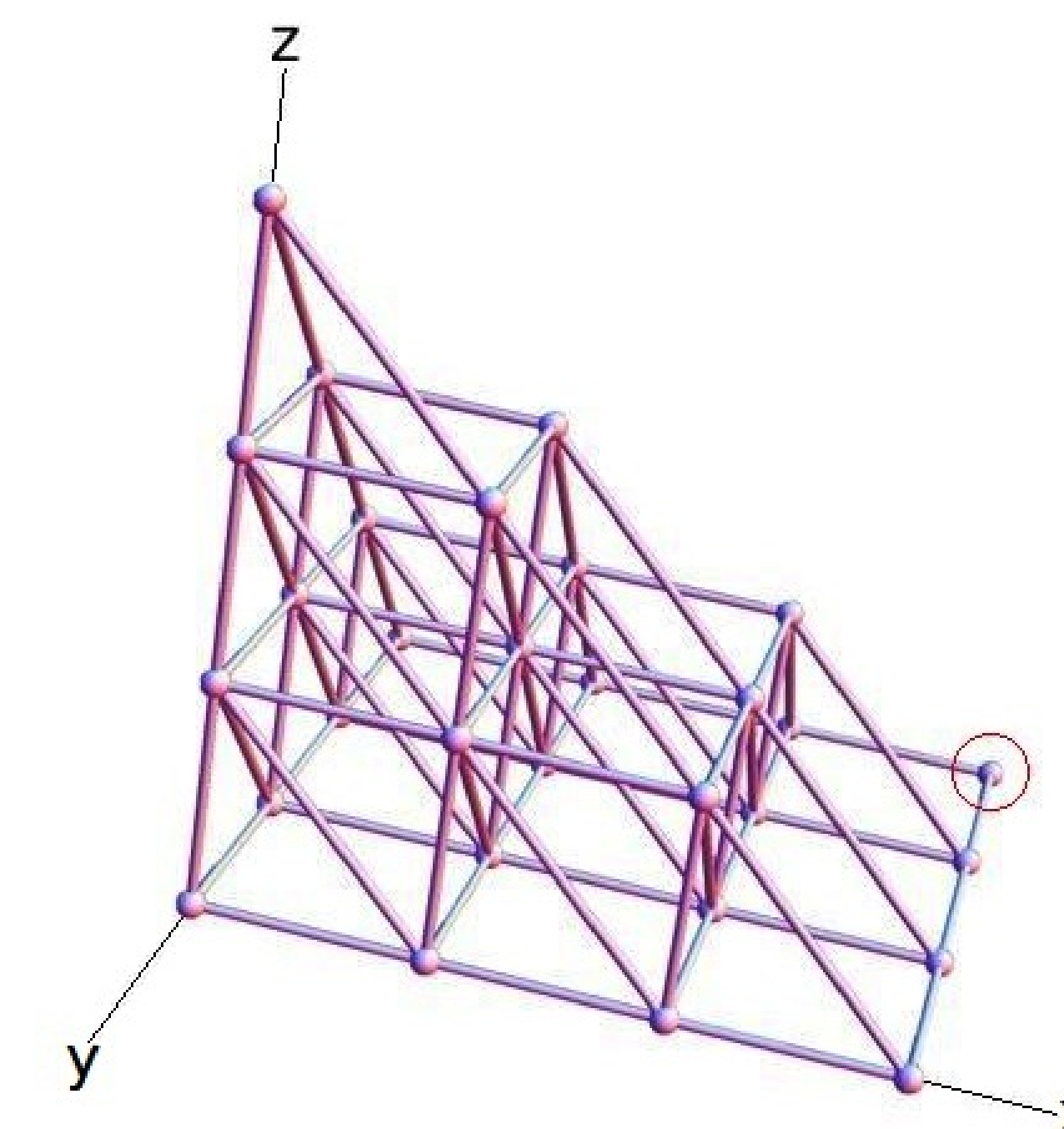
- The colors correspond to which superclass N is in. There are four such superclasses, namely

$$\begin{pmatrix} 0 & 0 \\ & 0 \end{pmatrix} \text{ (yellow), } \begin{pmatrix} 1 & 0 \\ & 0 \end{pmatrix} \text{ (green), } \begin{pmatrix} 0 & 0 \\ & 1 \end{pmatrix} \text{ (blue), and } \begin{pmatrix} 0 & 1 \\ & 0 \end{pmatrix} \text{ (red).}$$

- We denote the probability that N is in a superclasses by $P_{(0,0,0)}$, $P_{(1,0,0)}$, $P_{(0,1,0)}$, and $P_{(0,0,1)}$ respectively.
- The arrows correspond to the probability associated with the movement from one superclass to the another.
- The probabilities can be organized into a probability, or transportation, matrix which describes the random walk.

POLYTOPE

- The random walk can be viewed geometrically as a 3-dimensional polytope.
- The vertices correspond to the superclasses and the edges correspond to the possible moves between superclasses.
- Below depicts the case when $m = n = l = 3$.



- Conjecture:** The random walk will tend towards the furthest point from the origin on the xy-plane (circled in red).

PROBABILITY MATRIX

- The probability (transportation) matrix P is diagonalizable and $P = S^{-T} D S^T$ where S is the matrix representation of the supercharacter table.
- The eigenvalues of P are the diagonal entries of D . Let $D_{(a,b,c)}$ be a diagonal entry of D . Then

$$\begin{aligned} D_{(a,b,c)} = & P_{(0,0,0)} \\ & + q^\ell \frac{[n]}{[\ell]} (q^{2a}(q-1)[m-a][\ell-c-a] - [a]) P_{(1,0,0)} \\ & + \frac{q^m [l]}{[m]} (q^{2b}(q-1)[m-b][n-c-b] - [b]) P_{(0,1,0)} \\ & + \frac{[m]}{[n]} (q^{2c}(q-1)[n-c][\ell-c] - [c]) P_{(0,0,1)}. \end{aligned}$$

- The stationary distribution of P is determined by its eigenvalues. We look at $\lim_{n \rightarrow \infty} D_{(a,b,c)}^n$ to determine where the random walk will tend towards.